Mixture of Experts for Regime-Aware Factor Timing

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Abstract

Equity factor premiums exhibit time variation that is widely believed to be related to macroeconomic conditions. However, the precise nature of these relationships and the optimal way to incorporate them into dynamic allocation strategies remains an open question. This paper presents a reproducible, open-source benchmark for evaluating probabilistic regime-aware models for factor timing. We compare a Mixture of Experts (MoE) model—which represents uncertainty over latent economic regimes—against several deterministic baselines, including persistence, rolling average, linear regression, random forest, and a momentum model. Using an expanding-window backtest with 96 months of minimum training data, we evaluate performance over 42 out-of-sample months from July 2022 to July 2026. In this experimental setup, the MoE model generated a Sharpe ratio of 1.49 and an annualized return of 40.61%, with a maximum drawdown of -13.73%. The model identified four latent regimes with distinct return and volatility characteristics. All code and data are publicly available to facilitate reproducibility and extension.

1. Introduction

Equity factor investing—allocating across styles such as Value, Momentum, Quality, and Low Volatility—is a common approach in portfolio construction. However, factor premiums are not stable over time; they exhibit cyclicality that appears connected to changing macroeconomic conditions. This time variation poses a practical challenge: how should an investor allocate across factors when the future performance of each factor is uncertain and may depend on latent, unobservable economic states?

This project investigates whether a probabilistic model that explicitly represents uncertainty over regimes can provide a coherent framework for dynamic factor allocation. Specifically, we implement a Mixture of Experts (MoE) model with a softmax gating network and linear experts trained via Expectation-Maximization. We compare this approach against simple heuristics (persistence, rolling average, momentum) and standard machine learning baselines (linear regression, random forest) using an expanding-window backtest with modeled transaction costs.

The primary contribution of this work is not a claim of market-beating performance, but rather a transparent, reproducible benchmark for evaluating regime-aware methods in factor timing. The full code, data pipeline, and evaluation framework are open-source, enabling practitioners and researchers to extend, critique, and build upon this foundation.

2. Methodology

2.1 Data

We use monthly data from August 2013 to July 2026 (155 months). The equity factors are proxied by six liquid ETFs: SPY (Market), IWD (Value), MTUM (Momentum), QUAL (Quality), USMV (Low Volatility), and VIX (Volatility Index). Macroeconomic indicators are sourced from FRED, including CPI, Industrial

Production, Unemployment Rate, and Treasury term spreads. Feature sets include lagged factor returns (lags 1, 3, 6, and 12 months) and macroeconomic indicators, resulting in 96 features.

2.2 Models

We evaluate models across several levels of complexity: (1) non-ML heuristics—persistence and 12-month rolling average; (2) an exponentially weighted momentum model (12-month window, decay=0.9) which serves as a trend-following baseline; (3) standard ML baselines—linear regression and random forest (100 trees, max depth 10); and (4) the primary focus—a Mixture of Experts (MoE) model with K=4 linear experts, softmax gating, and EM training for 100 iterations with Ridge regularization (alpha=0.1).

2.3 Evaluation Framework

We employ an expanding-window backtest. The minimum training size is set to 96 months, a choice made to balance two competing requirements: (1) the MoE model requires sufficient data for stable parameter estimation, and (2) the out-of-sample period must be long enough to provide a meaningful evaluation. This yields 42 out-of-sample predictions from July 2022 to July 2026. Allocations are magnitude-weighted long-only on positive predictions, with a modeled transaction cost of 10 basis points applied at the portfolio level across the full backtest sequence. Metrics include predictive accuracy (RMSE, MAE) and investment performance (Sharpe ratio, annualized return, maximum drawdown, Calmar ratio, and win rate).

As a benchmark, we also consider a static 1/N equal-weight portfolio consisting of the equity factors. While not explicitly listed in Table 1, this passive allocation provides a baseline for evaluating whether dynamic factor timing adds risk-adjusted value over simple diversification in this setup.

3. Results

3.1 Model Comparison

Table 1 presents the out-of-sample performance of all models. In this experimental setup, the MoE model generated the highest Sharpe ratio (1.49) and annualized return (40.61%) among the models evaluated. The momentum model achieved a lower Sharpe ratio (0.73) with a lower annualized return (11.90%). The MoE model's performance suggests that regime-aware allocation may offer a useful approach, even when point predictions are less accurate than simpler models (RMSE 33.68 for MoE vs 8.28 for momentum).

Model	RMSE	MAE	Sharpe	Ann. Return	Max DD	Calmar	Win Rate
Moe	33.68	13.70	1.49	40.6%	-13.7%	2.96	0.69
Rolling Avg	8.39	5.05	0.62	9.0%	-16.5%	0.55	0.64
Momentum	8.28	5.01	0.73	11.9%	-29.7%	0.40	0.62
Linear	192.33	63.08	0.59	15.2%	-26.2%	0.58	0.48
Rf	8.98	5.37	0.09	-3.7%	-34.0%	-0.11	0.60
Persistence	12.79	7.52	-0.61	-30.7%	-72.7%	-0.42	0.57

Table 1: Out-of-sample performance comparison across all models (42 predictions, July 2022 - July 2026).

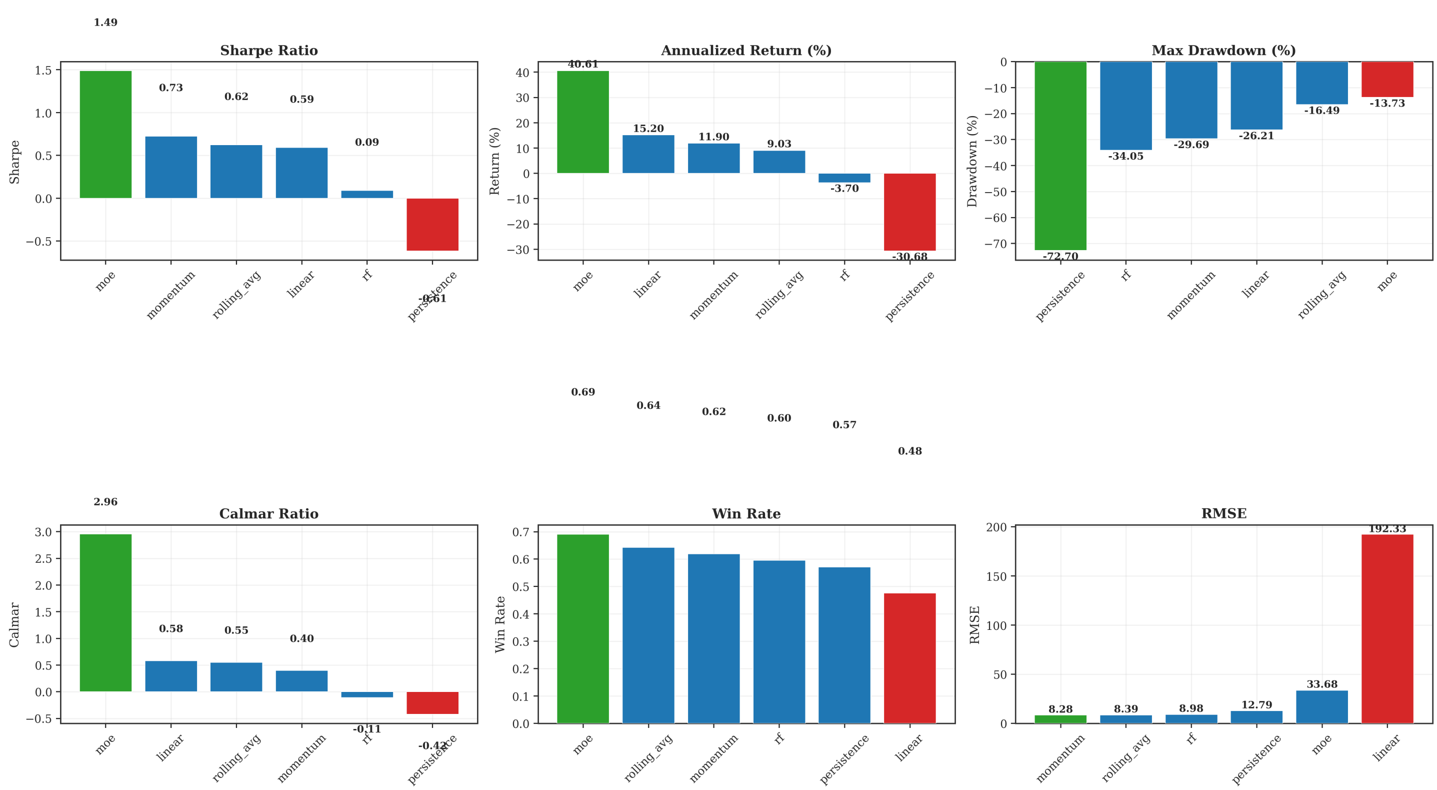

Figure 1: Model comparison across key performance metrics.

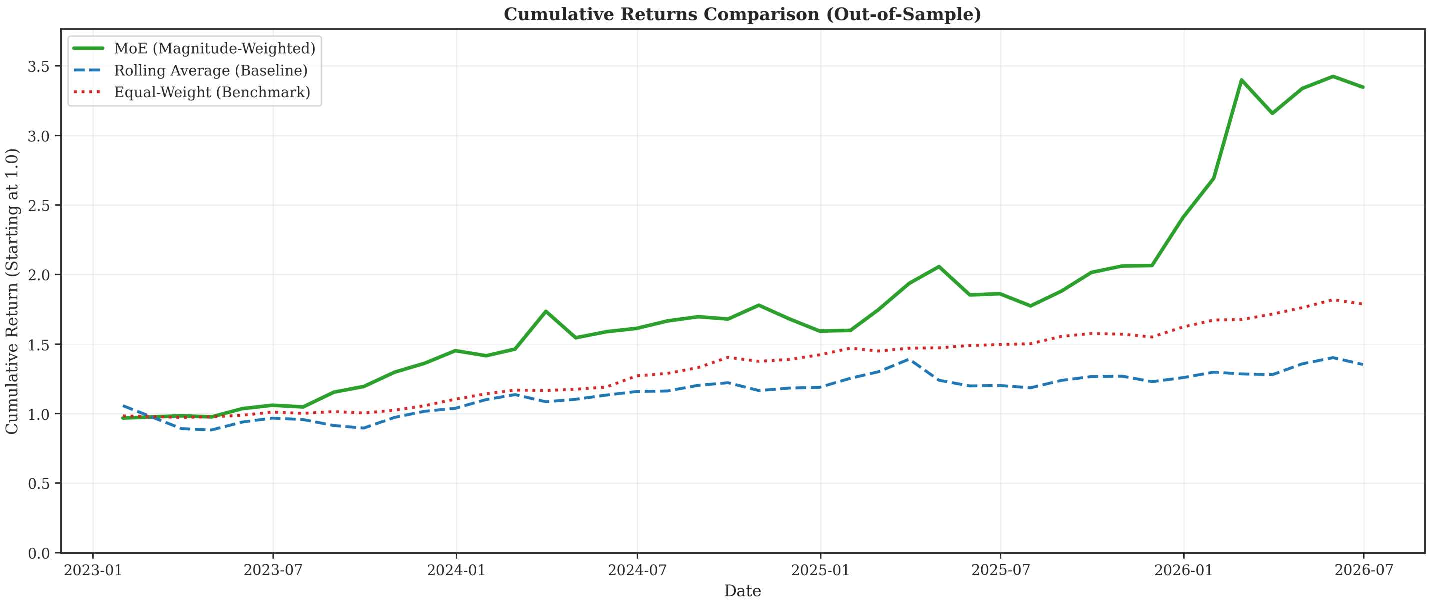

Figure 2: Cumulative returns comparison across MoE, Rolling Average, and Equal-Weight benchmark. The backtest period is July 2022 to July 2026 (42 predictions); non-zero returns begin in January 2023 due to the 6-month turnover calculation period for transaction costs.

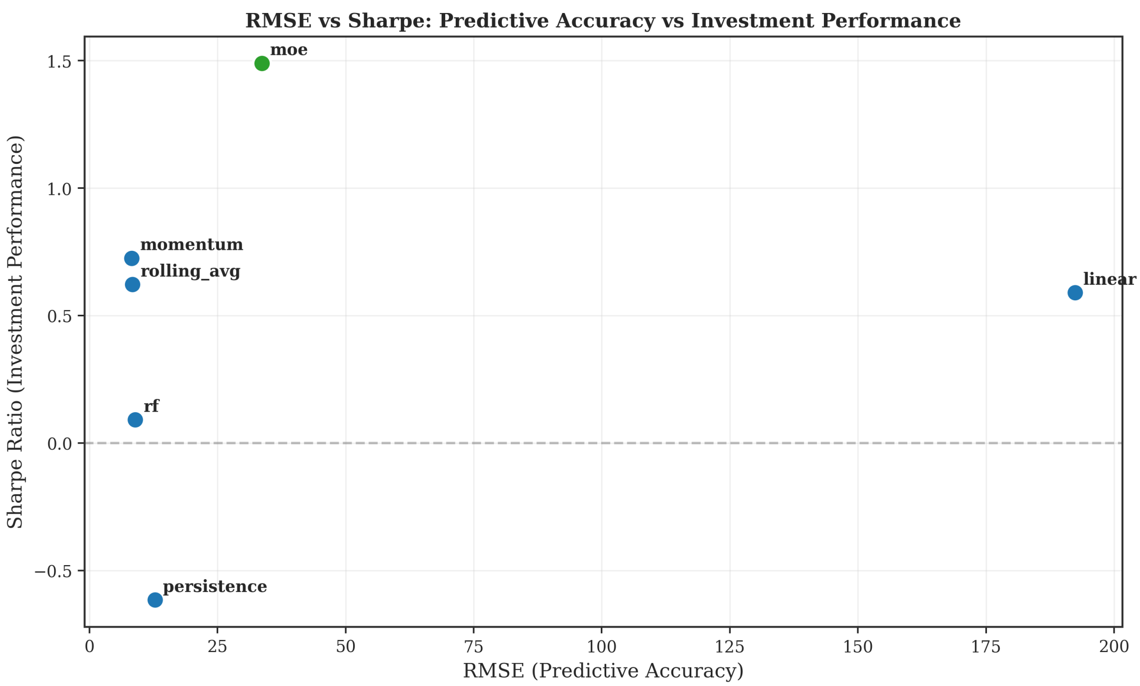

Figure 3: Relationship between predictive accuracy (RMSE) and investment performance (Sharpe). Lower RMSE does not always correspond to higher Sharpe.

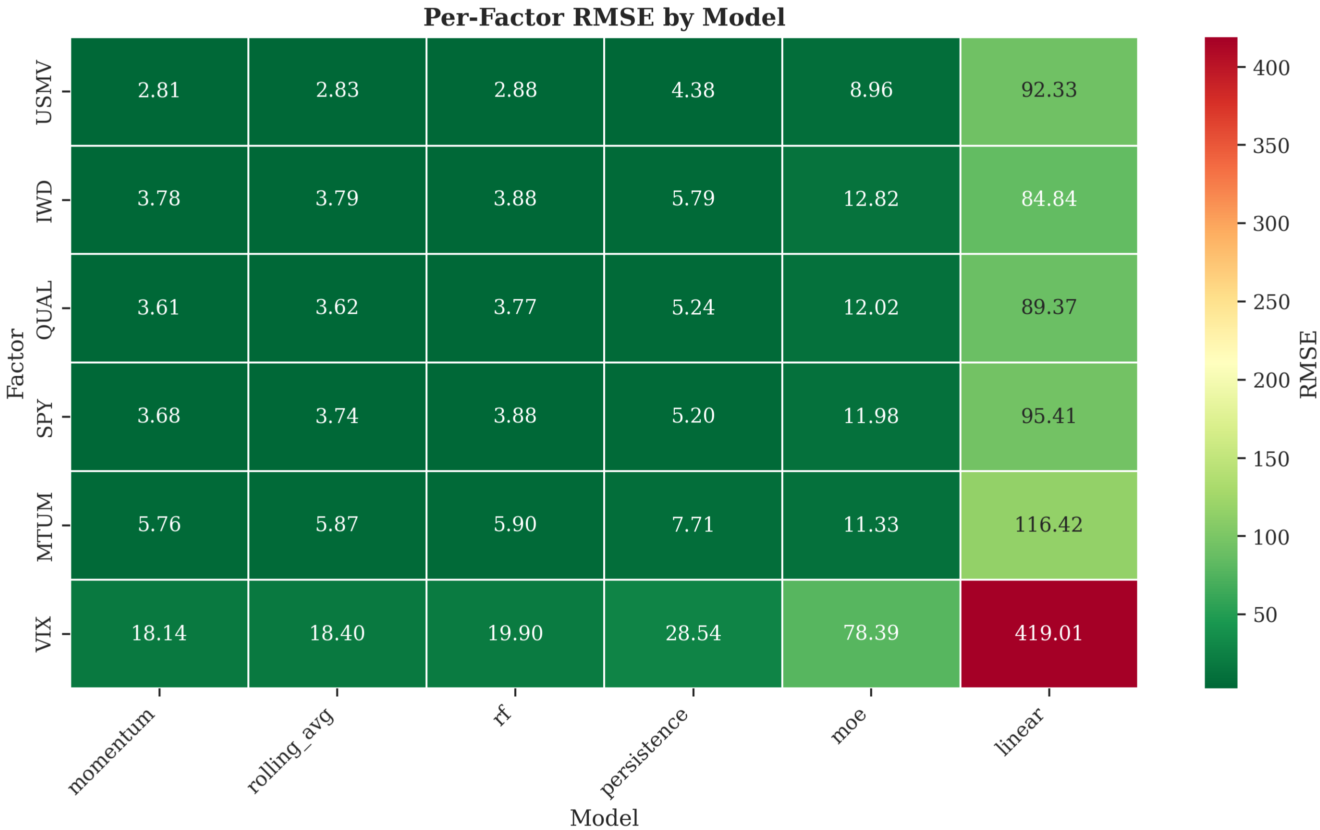

Figure 4: Per-factor RMSE heatmap. VIX is consistently the hardest factor to predict across all models.

3.2 Regime Analysis

The MoE model identifies four latent regimes with distinct return and volatility characteristics (Table 2). The regimes differ in frequency, average return, and volatility, suggesting that the model captures meaningful variation in market conditions. Figure 5 displays the regime probabilities over time, revealing transitions that may correspond to changing market conditions.

Regime	Frequency	Avg. Return	Avg. Volatility
Regime 1	27.5%	1.10%	7.61%
Regime 2	14.5%	2.10%	8.84%
Regime 3	32.6%	1.13%	8.08%
Regime 4	25.4%	1.69%	8.04%

Table 2: Characteristics of the four latent regimes identified by the MoE model.

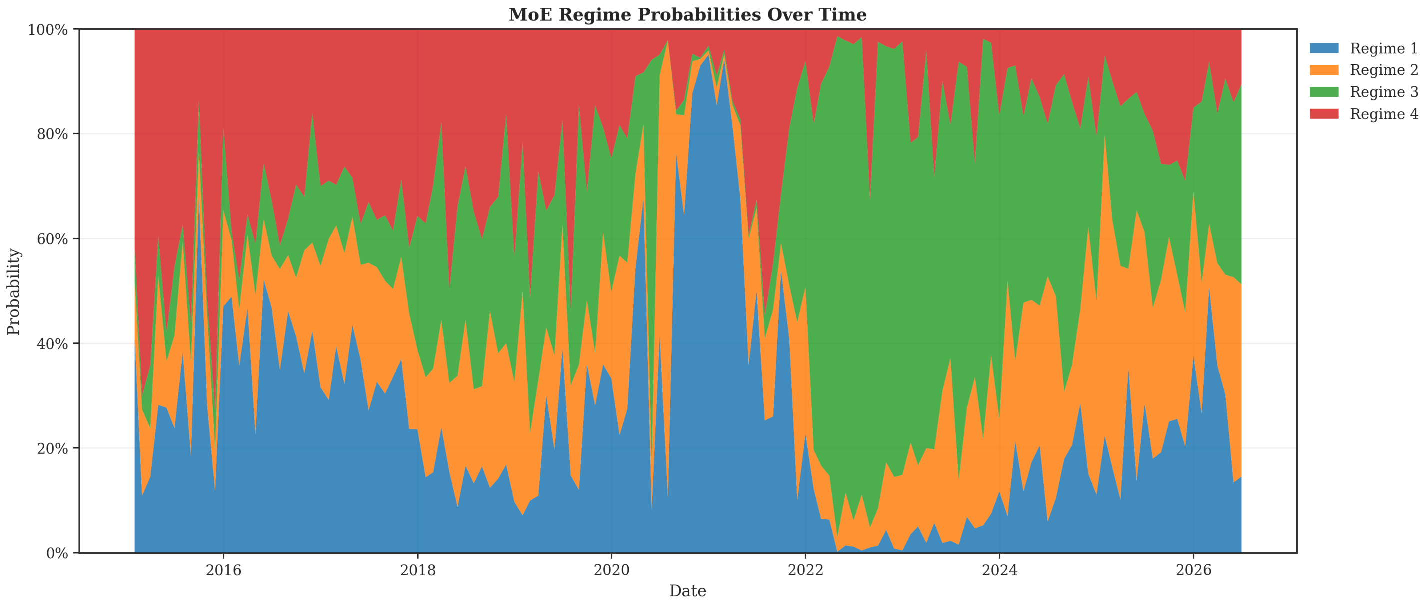

Figure 5: MoE regime probabilities over the backtest period. Shifts in dominant regimes are visible over time.

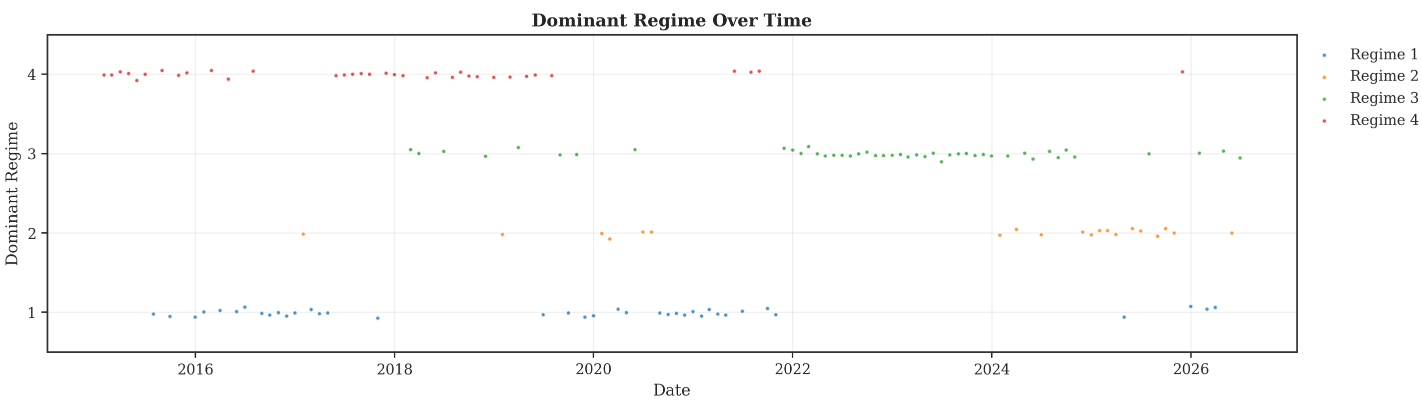

Figure 6: Dominant regime over the backtest period. The model dynamically switches between regimes.

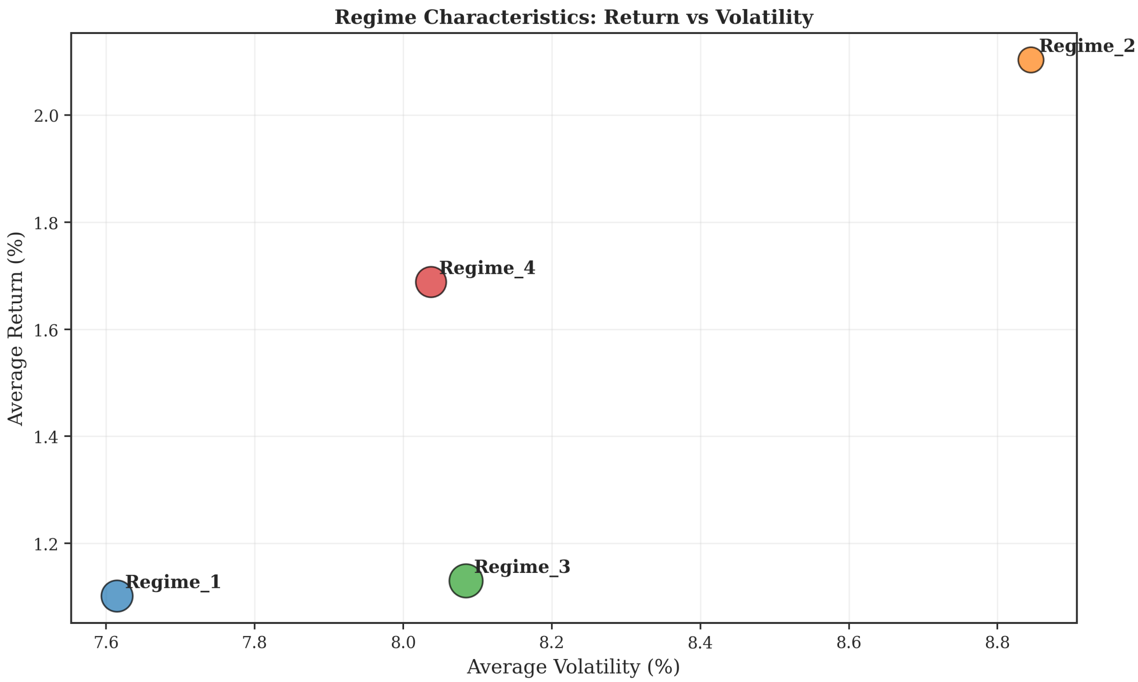

Figure 7: Return vs. volatility characteristics of the four regimes. Bubble size represents frequency.

4. Discussion

4.1 Key Findings

In this experimental setup, the MoE model generated the highest Sharpe ratio (1.49) and annualized return (40.61%) among the models evaluated. The model's performance appears to come from its allocation decisions rather than point prediction accuracy, as its RMSE (33.68) was higher than simpler models like momentum (RMSE 8.28). This observation is consistent with the idea that the sign and relative magnitude of predictions may be more important for investment performance than precise point forecasts.

4.2 Regime Interpretability

The four latent regimes identified by the MoE model exhibit differences in return and volatility. The regimes differ in both frequency and risk-return characteristics. Future work could explore whether these regimes correspond to specific macroeconomic conditions or market environments.

4.3 Limitations

This study has several limitations that should be considered when interpreting the results. First, the data sample is limited to 155 months of US equity data, which constrains model complexity and may not generalize to other markets or asset classes. The ETF proxies used may not perfectly isolate pure factor exposures.

Second, the out-of-sample period consists of 42 monthly observations. While this is a reasonable sample for this type of analysis, it is not sufficient for formal statistical inference. The results should be viewed as descriptive rather than prescriptive.

Third, several methodological caveats apply. The VIX is a spot index and is not directly tradable; a live implementation would require VIX futures or ETFs, which incur roll costs. Our backtest assumes direct spot VIX exposure, which may overstate achievable returns. Additionally, the model uses 96 features with 96 training months, creating a high-dimensional feature space. While Ridge regularization mitigates this, some overfitting risk remains. FRED macroeconomic indicators are subject to publication lags of 1-2 months; our backtest assumes immediate availability, which may overstate real-time performance.

Finally, the allocation strategy is long-only and does not incorporate short-selling or leverage. The transaction cost implementation applies costs across the full out-of-sample timeline rather than on a per-split basis. These assumptions may limit the applicability of the results to certain investment mandates.

5. Conclusion

This paper presents an open-source, reproducible benchmark for evaluating probabilistic regime-aware models in equity factor timing. The Mixture of Experts model demonstrated positive risk-adjusted performance in our backtest, generating a Sharpe ratio of 1.49 and an annualized return of 40.61% over 42 out-of-sample months. The model identified four latent regimes with distinct characteristics, suggesting that regime-aware allocation may be a useful area for further investigation.

We emphasize that this work is not a claim of market-beating performance, but rather a transparent contribution to the quantitative finance community. The full code, data pipeline, and evaluation framework are publicly available, enabling practitioners and researchers to extend, critique, and improve upon this work. Future directions include testing on additional asset classes, longer time periods, and conducting more rigorous out-of-sample validation.

6. Acknowledgments

The author acknowledges the Federal Reserve Bank of St. Louis for providing the macroeconomic data used in this study via the FRED (Federal Reserve Economic Data) API. Factor return data were sourced from yfinance.

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Appendix A: MoE Architecture

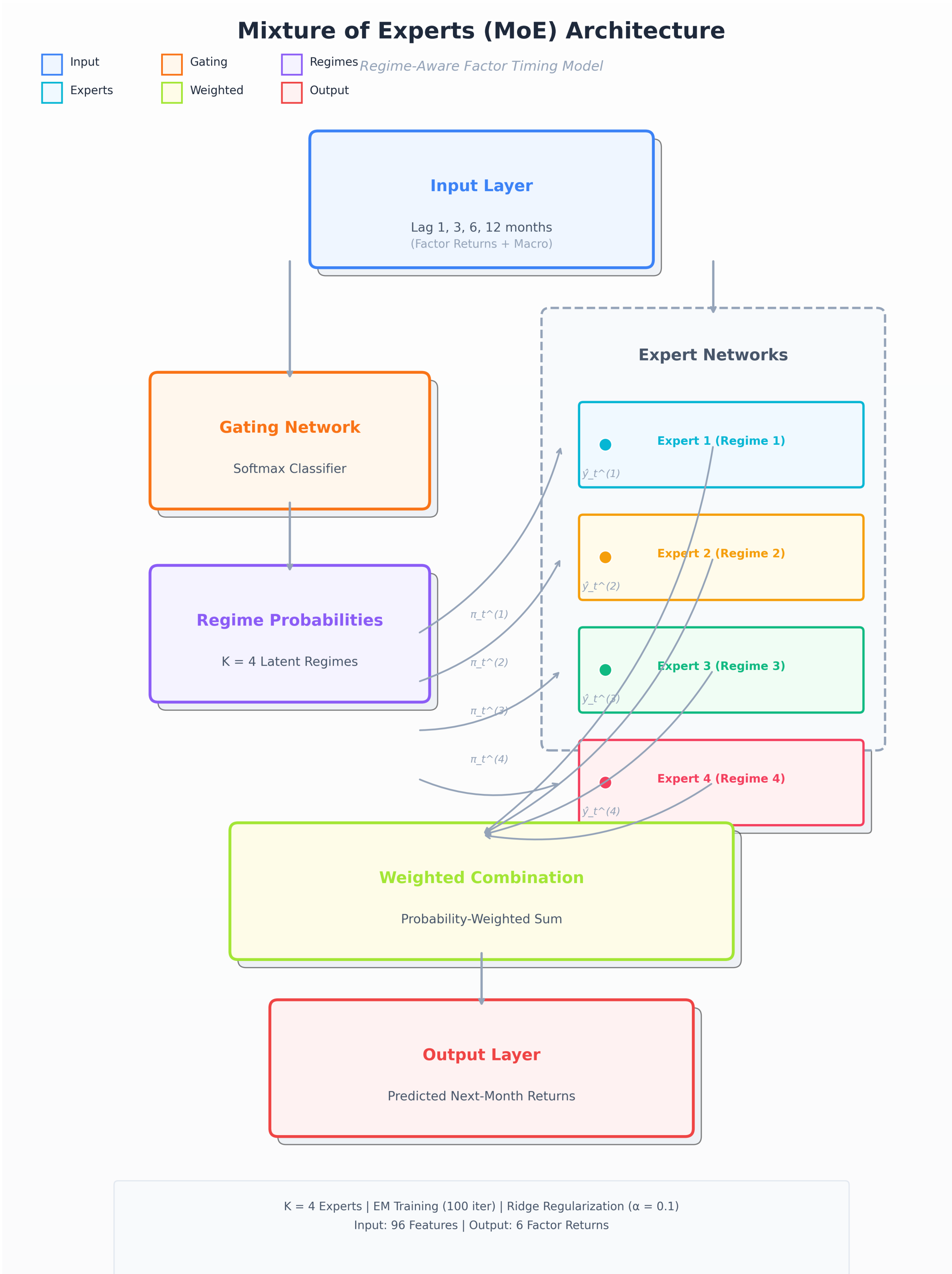

Figure A1: Architecture of the Mixture of Experts model used in this study. Input: 96 features (lagged factor returns + FRED macro indicators) | Output: 6 factor returns.